

ATAI Non-Helium VTOL Concepts

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1. Executive Overview

This document presents three conceptual engineering variants of the ATAI non-helium vertical takeoff and landing (VTOL) capsule platform. Unlike earlier hybrid buoyant concepts, these configurations rely entirely on thrust-based lift generated by distributed ducted fans. The goal of this concept study is to define realistic engineering envelopes for short-duration hover operations and short-range transit missions.

Primary mission requirements defined for this study: * Hover capability: 5 minutes continuous hover * Operational range: 10 km * Pure thrust lift (no helium or buoyant gas) * Multi-fan distributed propulsion architecture * High-stability landing system using 12 reinforced mechanical legs

Each concept integrates a heavy-duty landing platform consisting of twelve short mechanical legs with large-diameter springs and hydraulic shock absorbers. This system allows the vehicle to safely absorb landing loads, stabilize the craft during hover transitions, and support operation on uneven surfaces.

2. Mission Definition

The ATAI non-helium platform is envisioned as a tactical aerial capsule capable of rapid deployment, emergency extraction, and short-range aerial logistics.

Because the system does not rely on buoyant gases, it can operate in environments where helium logistics or large envelope structures are impractical.

Typical mission profile: * Vertical takeoff * Stabilized hover for up to 5 minutes * Controlled forward flight for 10 km * Precision landing on reinforced mechanical legs

Operational use cases include: * Tactical evacuation * Rapid cargo transport * Emergency rescue operations * Experimental aerospace platform testing

3. Core Platform Architecture

All three variants share the same fundamental architecture:

Distributed Electric Ducted Fans Multiple ducted fans arranged in a circular or radial configuration generate vertical lift and provide directional control.

Hybrid Power System Power is generated by a turbo-generator or turboshaft system paired with buffer batteries that provide peak power during takeoff and hover.

Structural Frame A reinforced composite and aluminum lattice frame supports the propulsion modules and payload capsule.

Landing System Twelve short mechanical landing legs equipped with heavy springs and hydraulic shock absorbers. These legs are designed to tolerate high vertical loads during landing.

4. Landing Platform Engineering

One of the defining elements of the ATAI system is the landing platform.

The landing structure consists of: * 12 reinforced mechanical legs * Short and wide geometry for maximum stability * High-strength steel or titanium shafts * Large mechanical springs for energy absorption * Hydraulic shock absorbers for damping oscillations

Benefits of this architecture: * High stability during landing * Ability to operate on rough terrain * Load distribution across 12 points * Reduced structural stress on the central frame

5. Variant Concepts

Three non-helium variants were developed for this engineering study.

Variant A ATAI NH-20 Scout

A compact tactical platform optimized for agility and rapid deployment.

Approximate specifications: * Diameter: 20 m * Propulsion: 10 ducted fans * Fan diameter: 2.2 m * Operational mass: 12 tons * Hover power requirement: 5 MW * Energy system: Turbo-generator + lithium buffer battery

Capabilities: * Hover time: 5 minutes * Range: 10 km * Crew capacity: 2-4 persons

Variant B ATAI NH-35 Tactical Carrier

A medium platform designed for cargo and rescue missions.

Approximate specifications: * Diameter: 35 m * Propulsion: 14 ducted fans * Fan diameter: 3 m * Operational mass: 28 tons * Hover power requirement: 12 MW * Energy system: dual turboshaft generators

Capabilities: * Hover time: 5 minutes * Range: 10 km * Payload capacity: ~6 tons

Variant C ATAI NH-60 Heavy Platform

A large heavy-lift concept designed for strategic payload transport.

Approximate specifications: * Diameter: 60 m * Propulsion: 18 large ducted fans * Fan diameter: 5 m * Operational mass: 65 tons * Hover power requirement: 28 MW * Energy system: four turboshaft engines driving distributed generators

Capabilities: * Hover time: 5 minutes * Range: 10 km * Payload capacity: 15+ tons

6. Comparative Specification Table

Platform	Diameter	Fans	Fan Diameter	Operational Mass	Hover Power	Range	Hover Duration	Landing Legs	Primary Energy
ATAI NH-20 Scout	20 m	10	2.2 m	12 t	5 MW	10 km	5 min	12	Turbo-generator hybrid
ATAI NH-35 Tactical	35 m	14	3 m	28 t	12 MW	10 km	5 min	12	Dual turboshaft hybrid
ATAI NH-60 Heavy	60 m	18	5 m	65 t	28 MW	10 km	5 min	12	Quad turboshaft system

7. Power System Considerations

Pure battery systems are currently insufficient for vehicles in the tens-of-tons class performing sustained hover. Therefore hybrid architectures are favored.

Advantages of hybrid systems: * Higher energy density * Reduced battery mass * Improved endurance * Greater operational flexibility

Jet fuel energy density is roughly twenty times higher than modern lithium batteries, making turbine-assisted systems more suitable for heavy VTOL platforms.

8. Stability and Control

The distributed propulsion architecture allows thrust vectoring and differential power control across multiple fans.

This provides: * Roll control * Pitch control * Yaw control * Hover stabilization

Advanced flight control software would continuously adjust thrust distribution to maintain stability.

9. Structural Materials

Possible materials for structural components include: * Carbon fiber composite frames * Aluminum-lithium aerospace alloys * Titanium reinforcement joints * High-strength steel landing legs

These materials balance strength, durability, and weight efficiency.

10. Engineering Challenges

Key challenges for non-helium heavy VTOL platforms include: * Extreme power demand during hover * Thermal management of turbine generators * Noise and airflow turbulence * Mechanical stress during landing * Redundancy requirements for safety

These factors must be addressed during detailed engineering development.

11. Development Path

A realistic development path could follow three phases:

Phase 1 Subscale Prototype Testing propulsion distribution and hover control.

Phase 2 Medium Platform Demonstrator Development of the NH-20 or NH-35 prototype.

Phase 3 Heavy Platform Development Scaling the architecture to the NH-60 heavy system.

12. Conclusion

The ATAI non-helium VTOL concept demonstrates a scalable thrust-based architecture capable of supporting short hover operations and short-range flight without the use of buoyant gases.

Three scalable variants NH-20, NH-35, and NH-60 illustrate how the system could evolve from a compact tactical platform to a heavy strategic aerial vehicle.

The use of distributed ducted fans, hybrid turbine power systems, and a reinforced 12-leg landing platform forms the core engineering philosophy of the ATAI non-helium design approach.

Further engineering work would include aerodynamic simulations, structural stress analysis, and prototype testing to validate these concepts.